Angudubo, S[1]., Benavente, G[2]., van Etten, J[2]., & Sinyinda, L[3]. (2026). Insights into the costing of on-farm testing: A case study of maize tricot trials by Zambia Agricultural and Research Institute (ZARI).

[1]Alliance of Bioversity International and CIAT, Uganda
[2]Alliance of Bioversity International and CIAT, France
[3]Zambia Agricultural Research Institute

1000FARMS

Insights into the costing of on-farm testing: A case study of maize tricot trials by the Zambia Agricultural and Research Institute

Angudubo, S., Benavente, G., van Etten, J., & Sinyinda, L. (2026).

Introduction

Costing on-farm testing

On-farm testing is essential for ensuring that agricultural innovations deliver value under the conditions where they are actually used. Performance observed on research stations often reflects optimal management, controlled environments, and uniform plots. In contrast, farmers operate under heterogeneous soils, microclimates, input constraints, labor trade-offs, and diverse market incentives. If varietal and technology evaluation does not involve these realities, recommendations risk being technically sound but contextually misaligned. On-farm testing, therefore, strengthens external validity: it captures genotype × environment × management (GxExM) interactions, reveals trait trade-offs that matter to farmers, and generates evidence that is directly relevant for adoption decisions. It also enhances legitimacy, as farmers become co-evaluators rather than passive recipients of innovation.

On-farm testing in African public breeding program have rarely performed on-farm testing of breeding products at scale, gathering sufficient data to make robust recommendations for variety release and promotion. To address this situation, the 1000FARMS platform was launched in 2021 with funding from the Gates Foundation and is implemented through partnerships led by the Alliance of Bioversity International and CIAT, together with IITA, CIMMYT, INN University (until 2023), and Wageningen University. The project works closely with CGIAR-NARES breeding networks and links to other CGIAR-led initiatives to support national testing and scaling efforts. The 1000FARMS project aims to establish an Africa-wide network for on-farm testing for better breeding decisions and variety placement recommendations.

One limitation in conventional ways to execute on-farm testing is their scalability. The tricot (triadic comparison of technologies) approach provides a scalable and statistically rigorous way to implement decentralized on-farm testing. Instead of testing many varieties in a few locations, tricot distributes small, incomplete blocks—typically three options—to many farmers. Each farmer ranks performance under their own management, while advanced statistical models aggregate these comparisons to estimate overall and segment-specific performance. This design reduces logistical complexity, increases geographic coverage, and allows credible inference across the target population of environments. By combining farmer-generated data with robust analytics, tricot bridges scientific rigor and real-world relevance, strengthening variety release, product design, product placement, and seed system decisions.

Scalability also depends on the cost per data point of on-farm testing. Costs of on-farm testing depend much on the operational decisions made by the implementers. To better understand what drives the costs of on-farm testing and to refine our costing methodology, we conducted a series of case studies. This study focuses on maize in Zambia.

Case study context: ZARI

The Zambia Agricultural Research Institute (ZARI) is one of the national institutions collaborating with the Alliance of Bioversity International and CIAT, particularly through the 1000FARMS platform project via the CGIAR-NARES network. ZARI operates as a department under Zambia's Ministry of Agriculture (MoA), with a core mandate to conduct agricultural research. Its major focus is to generate agricultural

production technologies suitable for all farmer categories and varying farming environments. The institute achieves this mandate through multiple research divisions, including crop improvement and agronomy, with its maize research program forming a significant part of these efforts.

Within its maize research program, ZARI is responsible for testing and advancing candidate maize varieties based on performance, farmer preference, and suitability across real farm conditions. For this specific maize trial, the tricot approach was chosen because it enables ZARI to efficiently collect structured feedback from farmers and collect field performance data on a large scale. The tricot trial is implemented using a practical model that relies on local facilitation, primarily by extension staff or lead farmers, ensuring effective delivery and engagement at the community level.

This report analyzes the costing of ZARI's maize tricot trial, focused on an intermediate-maturity product profile. The trial compares eight technology options across 110 participating farmers (blocks) in a full cycle from November 21, 2024, to April 1, 2025 (131 days after planting). Field agents carry out routine monitoring through four visits, while the research team conducts two additional visits for supervision and quality checks. ZARI coordinates seed distribution centrally, and local field agents manage last-mile delivery so that all farmers receive seed on time. ZARI implemented the trials in nine districts: Chibombo, Chisamba, Kabwe, Kapiri Mposhi, Lusaka, Kafue, Mazabuka, Chipata, and Mansa, across Central, Lusaka, Southern, Eastern, and Luapula Provinces, covering three agro-ecological regions. The team used digital data collection, which strengthened the capacity of the maize program by enabling technicians to submit trial data more promptly and consistently.

This report presents the costing methods and key assumptions in the estimation. It outlines the major expenditure components and shows how each contributes to total implementation costs. It highlights the major cost drivers and shows how deviations from the tricot steps affect efficiency. The report also provides an overview of what the results mean for variety development and scaling of on-farm testing. Finally, it offers recommendations to improve cost-efficiency and support institutional integration of decentralized on-farm testing.

Costing objectives

Systematic costing expresses trial implementation resource needs as unit costs per data point (farmer or block). But the benefit of costing goes beyond obtaining the numbers. Costing helps programs understand how operational decisions, such as who does what, where, and how often, translate into resource requirements. These operational decisions have consequences for data quality or farmer engagement. Therefore, by having clarity about costs and trade-offs with other dimensions, programs can analyze these operational decisions in a way that is transparent and comparable across partners and locations. Such comparisons help programs to learn about these trade-offs and to better plan their scaling efforts. In summary, costing helps with the following objectives.

1. Supporting realistic budgeting and resource mobilization;
2. Identifying efficiency gains without compromising data quality or partnership quality;
3. Enabling fair and sustainable cost-sharing among research, extension, and seed system stakeholders; and
4. Allowing for learning and improving delivery models across institutions and regions.

Costing methodology and assumptions

The tricot costing is prepared using the activity-based Excel Costing Template (https://hdl.handle.net/10568/182169) following the procedures outlined in the "Tricot Costing Template Guidelines" (https://hdl.handle.net/10568/181933). In particular, the guidelines specify that:

- Travel expenses are estimated using United Nations travel reimbursement standards (https://hr.un.org/node/53991).
- Communication costs follow country-level ICT price baskets as published by the International Telecommunication Union (ITU) (https://www.itu.int/en/ITU-D/Statistics/Dashboards/Pages/IPB.aspx).
- Unit prices for materials and services are based on market prices in each country and are adjusted for inflation where needed.

The key assumption is that the maize breeding program implemented tricot trials as prescribed, following the ten steps outlined in "The Tricot Guide" (https://hdl.handle.net/10568/109942).

Other assumptions include:

- The research team leads the initial work: trial preparation, experimental design, and seed packaging.
- Farmer-facing organizations handle most implementation activities because they work directly with farmers. The research team focuses on quality assurance through spot checks to support trial implementation under real farm conditions.
- Field agents from farmer-facing organizations receive sufficient training and then oversee field implementation on their own.
- ZARI plans seed production for tricot trials separately from other breeding trials to ensure the required quantities are available on time.
- The research team distributes seed centrally to agreed drop-off points, and field agents manage the last-mile delivery to participating farmers.
- Field agents lead all monitoring and data collection. The research team supports specialized tasks such as yield measurement and final farmer feedback workshops.
- The central breeding budget covers the research team's time for trial preparation, seed packaging, and training of field agents. While 10 – 15% share of the staff costs of researchers is allocated to on-farm testing.
- The costing uses institution-specific daily and hourly labor rates, calculated from official monthly salary scales for each job role.

Cost structure and components

This section summarizes the cost components and provides insight into the stage-level and role-level cost patterns. The findings on the cost structure are given in Table 1.

Table 1: Tricot implementation by component

Cost component	Sub-category	Trial cost (USD)	Share of total cost (%)	Cost/ farmer (USD)
Travel expenses	Per diem	5,752	36.8	52.29
	Travel allowance	764	4.9	6.95
Material	-	4,474	28.6	40.67
Labor	Field-based	2,236	14.3	20.33
Transport	Fuel & O/M	1,445	9.2	13.13
Communication	-	973	6.2	8.84
Total		**15,644.23**		**142.22**

Key observations

Table 1 summarizes the marginal (variable) implementation costs for the ZARI maize tricot trial. These costs increase as ZARI recruits more farmers because they relate to field delivery. In contrast, office-based activities count as fixed costs. The ZARI team incurs these costs to set up and coordinate the trials, and they do not change as the number of farmers increases. These costs cover meetings for the selection of test lines and traits for evaluation, planning for trial establishment, trial design, testing, and deployment on ClimMob, and centralized training of field agents organized by the breeding program.

Research staff time for office-based trial preparation falls under fixed costs. If the maize program also conducts training of field agents at the institute (the trial coordination center), this also remains a fixed cost because the program organizes the training once for the whole network, not each farmer. Under that option, total fixed costs increase. In this analysis, fixed costs were not considered.

The summary of ZARI maize tricot costing shows that travel expenses, materials, and field-based labor are the largest cost components. Together, they total USD 13,227, representing approximately 85% of total costs. The breakdown of these components is presented below:

- **Travel expenses:** Travel expenses, composed of per diem and travel allowances, are the largest cost component, accounting for 42% of total costs. Travel by the research team makes up approximately 61% of total travel expenses, indicating substantial researcher involvement in routine field activities, especially during monitoring and data collection.

- **Material costs:** Materials are the second largest cost component, accounting for approximately 29% of total costs. They include trial seeds, implementation supplies, and items supporting farmer participation. In the ZARI maize tricot costing, seed production accounts for about 35% of material costs, but only 10% of total costs. Seed production, therefore, does not drive total costs, but remains essential because trial success depends on timely access to quality seed.

  The remaining 65% of material costs support trial implementation. Participant support, mainly transport reimbursements and refreshments, takes the largest share of about 59% of material costs, making it the most significant line item. Participant support costs stay high when the

program runs many separate workshops and does not cluster farmers for joint sessions, because each additional session adds more transport and meal costs.

- **Labor costs:** Field-based Labor is the third largest cost component, accounting for 14% of total costs. It includes time spent during trial preparation, training of field agents, seed distribution, monitoring, data collection, yield measurement, and farmer feedback. Monitoring constitutes the largest sub-category within field-based labor costs (85% of labor costs), with a significant portion (about 48%) attributed to two research-team visits covering approximately 25% of participating farmers. Monitoring-related labor costs are high because of the involvement of multiple research roles, including associate breeders, research officers, research assistants, and technicians, rather than relying solely on local field agents for routine follow-up.

- **Transport costs:** Direct transport expenses (fuel and vehicle operation/maintenance) account for a relatively small share (9.2%) of total costs. Most transport spending is linked to field activities that require travel to many farmers, especially seed distribution, extension-led monitoring, and data collection. Distances are higher during seed distribution because extension staff collect seed from ZARI and deliver packages to farmers in their operation area. Transport needs also increase when field agent training is conducted across multiple locations. In contrast, transport costs for research-team monitoring and yield measurement are lower because the research team covered only approximately 25% of participating farmers' plots.

- **Communication costs:** Communication accounts for only 6.2% of total costs. It supports coordination and routine farmer follow-up. In the ZARI maize tricot costing, communication is estimated as a standard monthly bundle per job role, based on a low-consumption basket of 70 call minutes, 20 SMS, and 500 MB of data per month. Total communication cost depends on the number of staff and the trial duration. Extension staff account for nearly 70% of communication costs because they are the main link with farmers and therefore require more bundles.

Cost drivers in the maize tricot implementation

Tricot trial costs are shaped by operational decisions, including how activities are designed, scheduled, and scaled. In the ZARI maize tricot trial, travel time, materials, and labor time constitute the largest share of the total implementation costs.

In the ZARI maize tricot trial, the first cost driver is an operational decision to assign multiple research roles to routine field activities. Researchers carried out two monitoring visits and metric data collection for about 25% of participating farmers. In practice, this creates duplicated field presence: researchers and local field agents travel to the same trial sites to carry out overlapping tasks. Costs rise because each visit triggers per diem, travel allowances, and staff time across multiple roles, without a clear improvement in the quality of evidence.

This is most visible in travel expenses, which are the largest cost component. Because researcher travel is financed through per diem and travel allowances, each additional trip quickly increases total costs, even when fuel and vehicle maintenance remain modest. A more cost-effective option is to delegate routine monitoring and data collection to trained local field agents and refocus researchers on quality assurance activities, including planned spot checks and remote verification of submitted data. With adequate training and clear standard operating procedures (SOPs), this local delivery model can maintain oversight and data quality while reducing travel expenses by about 24%.

Labor costs follow the same logic. They are driven not only by the number of activities, but also by who performs routine tasks and how often they travel. When multiple research roles take part in routine visits, paid staff-days accumulate quickly, even when the tasks per farmer are simple and repeatable. Shifting routine monitoring and data collection for the 25% sample to trained local field agents reduces the need for routine researcher field time and focuses researcher effort on quality assurance. Under the costing assumptions, this operational shift would reduce field-based labor costs by 50.50%, while maintaining technical oversight and avoiding high labor costs driven by routine researcher travel.

A second cost driver is participant support, mainly transport reimbursements and refreshments. In the ZARI maize tricot costing, participant support accounts for 59% of material costs and about 17% of total costs, equivalent to approximately USD 24 per farmer. These costs are high not because the support items are expensive, but because the delivery model relies on many small, decentralized meetings for training, seed distribution, and farmer feedback. Each meeting adds transport reimbursements and refreshments and can also increase travel needs for the implementing team.

At the same time, participant support matters for participation and retention in a citizen-science model, such as tricot, where farmers contribute time and information voluntarily. The operational challenge is to provide enough support to keep farmers engaged, without allowing costs to rise with every additional meeting. A more cost-effective option is to organize activities through cluster hubs (for example, 10 – 20 farmers per hub) and bundle activities so that training, seed distribution, and feedback are delivered in fewer sessions. In the scenario analysis, a 30% reduction in transport reimbursements and refreshments reduces material costs by about 18%. Larger reductions (e.g., 50% or 70%) increase savings but also increase the risk of lower participation if not managed carefully. The key is to reduce the number of reimbursement points through better meeting design, while keeping support modest, transparent, and aligned with the goal of completing the trial.

Finally, costing on-farm testing requires a clear distinction between trial delivery and routine breeding work. One point of discussion is how much of the staff costs of the researchers are allocated to the on-farm testing. For the maize tricot trial, it is reasonable to treat office-based activities, such as trial preparation and training of field agents, as part of the core breeding budget, especially when coordinated at ZARI as the trial hub and not requiring travel. On-farm testing would then carry a small share of researcher's time to cover essential project-supported supervision (e.g., 10% as proposed by the ABI Transform team or 15% as applied in the ZARI maize tricot costing). The remaining staff time is budgeted centrally as routine breeding effort. This does not change field implementation costs, but it improves how budgets are interpreted and compared, reduces double-counting, and clarifies what projects are financing.

The scenario analysis shows that costs in the ZARI maize tricot trial are largely shaped by a small set of operational choices. When routine fieldwork depends on researchers' travel and many small meetings, costs rise quickly as farmer numbers increase. When delivery shifts to local implementation, supported by targeted quality assurance, costs fall, and the model becomes easier to scale.

If routine monitoring and data collection are delegated to trained local field agents, travel expenses decline. Under this scenario, the cost per farmer falls from USD 142.22 to USD 128.13 and total costs drop by 10%, saving USD 1,550. With the same operational shift, labor costs are also reduced. The cost per farmer falls from USD 142.22 to USD 132.16, and total costs fall by 7.1%, saving USD 1,106. This message here is that ZARI can expand to more farmers without a proportional increase in costs, because routine delivery no longer depends on repeated researcher travel and accumulated staff-days.

Participant support shows a different pattern. A 30% reduction in transport reimbursements and refreshments, achieved through fewer meetings and lower reimbursement levels, lowers the cost per

farmer to about USD 135, reduces total cost by about 5%, and saves USD 795. Larger reductions generate more savings (USD 1,324 at 50% or USD 1,854 at 70%), but they also increase the risk of lower participation and incomplete trials if support becomes too low. Participant support is therefore an operational design choice: it can be reduced, but it is best managed through clustering and bundling of meetings, rather than reducing support in a way that undermines retention.

Taken together, the scenarios show that scaling tricot is mainly a matter of delivery design. If ZARI adopts local delivery as the default and uses clustered engagement to manage participant support, the program can reduce the cost per farmer to about USD 103, saving roughly USD 4,270, and expand coverage while maintaining data quality and completing the trial cycle.

Conclusions and recommendations

Conclusions

The ZARI maize tricot costing shows the trial is feasible and can generate useful evidence. The main limitation is the current operational decisions. Costs are driven less by the amount of data collected and more by how implementation is organized, especially when routine tasks lead to repeated field trips (especially by the research team) and many small meetings. Where delivery is streamlined and roles are clear, ZARI can achieve the same trial outputs at a cost that the breeding program can sustain as the number of farmers increases.

Recommendations

Adopt a "no duplication" delivery model with audit-based supervision: ZARI should clearly separate routine delivery tasks, such as monitoring and data collection, which trained local implementers can handle, from researcher roles focused on quality assurance, including planned spot checks and remote verification of submitted data. This keeps scientific oversight strong while preventing routine delivery from becoming travel-heavy and makes costs more predictable as the number of farmers and coverage expands.

Cluster farmer engagement and define a clear support package: ZARI should organize farmers through cluster hubs and bundle key activities to limit the number of meetings. Transport reimbursements and refreshments should follow a simple, agreed package that is transparent and easy to apply. Adjustments to participant support should be guided by participation levels and trial completion risk, not ad hoc requests. This will improve trial completion rates and reduce missing data, strengthening the evidence base for robust variety recommendations.

Separate delivery costs from core breeding costs: ZARI should budget office-based trial preparation and centralized training of field agents under the maize breeding program central budget, and charge on-farm testing only for the supervision share of researcher time linked to field delivery. This improves transparency, avoids double-counting, enhances comparability across seasons, and supports more stable financing beyond project cycles.

Institutionalize the delivery model: ZARI should update SOPs and role guidelines to reflect the modified delivery rules, and monitor a small set of operational indicators, including cost per farmer, completion rates, and audit findings, to confirm that cost reductions translate into reliable evidence for variety release and scaling. This will create a repeatable system that can be applied across seasons and locations, and build confidence among partners and regulators in participatory, decentralized maize on-farm testing as part of the national variety release and scaling pathway.

Acknowledgements

I thank Rachel Chase for the support in developing the tricot costing template and for the feedback that strengthened this report. I also thank Lubasi Sinyinda for coordinating field implementation and for supporting the costing and data inputs from the ZARI maize program. I am grateful to the ZARI maize technicians and field agents for their time, records, and practical insights, which made it possible to prepare this report.

Suggested citation:
Angudubo, S., Benavente, G., van Etten, J., & Sinyinda, L. (2026). *Insights into the costing of on-farm testing: A case study of maize tricot trials by the Zambia Agricultural Research Institute (ZARI).* Alliance of Bioversity International and CIAT, Kampala, Uganda